ECA0806-Analog & Digital Communication

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Experiment-9

PERFORMANCE EVALUATION OF AM GENERATION AND DETECTION UNDER NOISY CHANNEL CONDITIONS USING MATLAB

Program (Matlab Code):

```
% Experiment 9: Performance Evaluation of AM under Noisy Channel (Toolbox-Free)
clc;
clear;
close all;

% Parameters
fs = 100000;          % Sampling Frequency (Hz)
t = 0:1/fs:0.02;      % Time Vector (s)
fm = 500;             % Message Frequency (Hz)
fc = 5000;            % Carrier Frequency (Hz)
SNR_dB = 10;          % Signal-to-Noise Ratio (dB)[cite: 51]

% Message Signal[cite: 51]
m = cos(2 * pi * fm * t);

% Carrier Signal[cite: 51]
c = cos(2 * pi * fc * t);

% AM Modulation[cite: 51]
am = (1 + m) .* cos(2 * pi * fc * t);

% Add AWGN Noise Manually (Bypasses Communications Toolbox 'awgn' function)
sig_power = mean(am.^2);
snr_linear = 10^(SNR_dB / 10);
noise_power = sig_power / snr_linear;
noise = sqrt(noise_power) * randn(size(am));
noisy = am + noise;

% Envelope Detection via Rectification and Low-Pass Filtering (Bypasses Signal Processing
Toolbox 'hilbert')
rectified = abs(noisy);
window_size = 40;
b_lpf = ones(1, window_size) / window_size;
env = filter(b_lpf, 1, rectified);
env = env - mean(env);                % Remove DC component
env = env / max(abs(env)) * max(abs(m));  % Scale amplitude for clear comparison
```

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```
% Display Results in Time Domain Subplots[cite: 51]
figure('Name', 'AM Performance under AWGN', 'NumberTitle', 'off');

subplot(5, 1, 1);
plot(t, m, 'LineWidth', 1.2);
title('Message Signal');
grid on;

subplot(5, 1, 2);
plot(t, c, 'LineWidth', 1.2);
title('Carrier Signal');
grid on;

subplot(5, 1, 3);
plot(t, am, 'LineWidth', 1.2);
title('AM Signal');
grid on;

subplot(5, 1, 4);
plot(t, noisy, 'LineWidth', 1.2);
title(['AM Signal with AWGN Noise (SNR = ', num2str(SNR_dB), ' dB)']);
grid on;

subplot(5, 1, 5);
plot(t, env, 'r', 'LineWidth', 1.5);
title('Recovered Signal');
grid on;
```

Output Waveform :

